YEAR 9 PHYSICS, AQA GCSE, UNIT 2

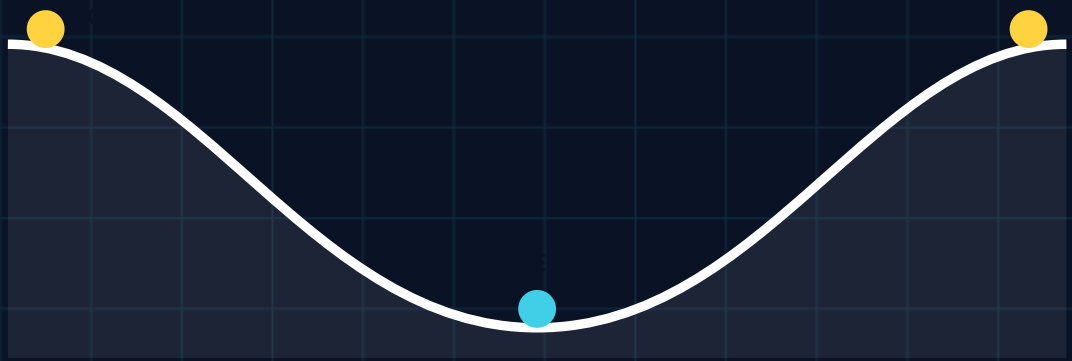
# ENERGY STORES AND TRANSFERS

Where energy is stored, how it moves, how fast it moves and how much of it is useful: nine lessons, from your first energy store to confident exam answers.

**9** LESSONS

**6** PRACTICALS

**8** EQUATIONS



NAME _____

CLASS _____

TEACHER _____

TARGET GRADE _____

START HERE

# HOW THIS BOOKLET WORKS

Energy is never made and never destroyed: it only moves from one store to another. This booklet shows you how to follow it, and how to calculate it. Keep the booklet: it is your classwork book *and* your revision guide.

**DO NOW**
Quick questions from earlier lessons. Remembering again is what makes it stick.

**LEARN IT**
The explanation, diagrams and worked examples. Read this again when you revise.

**TRY IT**
Practice that builds up: first with support, then on your own.

**PRACTICAL**
Record your measurements and work out results from them.

**EXAM CORNER**
AQA-style questions with a tip on how the marks are given.

**REVISE IT**
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: Energy is measured in _____ **(joules / newtons / watts)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of energy is | **the watt** | **the joule** | **the newton** | .

## THE ROUTE THROUGH THE UNIT

| # | LESSON | WEEK |
|---|---|---|
| **01** | **Energy stores and systems**<br>The energy stores, four ways to transfer energy, systems | 1 |
| **02** | **The kinetic store**<br>$E_k = \frac{1}{2} m v^2$; grams to kilograms; double the speed, four times the energy | 1 |
| **03** | **The gravitational potential store**<br>What $g$ means, $E_p = m g h$, height gained, finding $h$ or $m$ | 2 |
| **04** | **Stretching springs**<br>Two forces, elastic and inelastic, extension, $F = k e$, two graphs | 2 |
| **05** | **The elastic potential store**<br>$E_e = \frac{1}{2} k e^2$; two-step questions; double the stretch, four times the energy | 3 |
| **06** | **Conservation of energy**<br>Closed systems, falling and swinging, energy bar charts, dissipation | 3 |
| **07** | **Power**<br>Power as a rate, watts, $P = \frac{E}{t}$, how powerful are you? | 4 |

| # | LESSON | WEEK |
|---|---|---|
| 08 | **Wasted energy**<br>Dissipation, lubrication, thermal conductivity, insulation, design an Eco House | 4 |
| 09 | **Efficiency**<br>Efficiency from energy or power, decimals and percentages, Sankey diagrams | 5 |

The **equation toolkit** on the last page has every equation in the unit, with formula triangles to help you rearrange.

Lesson

# 01 ENERGY STORES AND SYSTEMS

AQA Physics 4.1.1.1; Combined Science 6.1.1.1

**BIG QUESTION** **A skateboarder rolls down one side of a ramp and up the other, but never quite gets back to the starting height. Where did the energy go?**

### BY THE END YOU CAN

- name the energy stores and give an example of each
- name the four ways energy is transferred
- explain what a system is
- describe the energy transfers when something changes

**energy store** **transfer** **system** **closed system** **joule (J)**

### YOU WILL NEED

This booklet, a pencil and a ruler.

REMEMBER

## WHAT DO YOU ALREADY THINK?

No wrong answers: this is your starting point.

**1**

Name something that has energy. How can you tell?

**2**

Energy is measured in _____ **(joules / newtons / metres)**.

**3**

Rub your hands together. They get _____ **(warmer / colder)**. Why?

EXPLAIN

# STORES AND TRANSFERS

Energy is never made and never used up. It is **stored**, and it is **transferred** (moved) from one store to another. Energy is measured in **joules (J)**.

## THE ENERGY STORES

| STORE | ENERGY IS STORED BECAUSE… | EXAMPLE |
|---|---|---|
| **Kinetic** | the object is moving | a rolling football |
| **Gravitational potential** | the object is raised up | a book on a high shelf |
| **Elastic potential** | the object is stretched or squashed | a stretched spring, a drawn bow |
| **Thermal** | the object is hot | a cup of tea |
| **Chemical** | of the substances it is made of | food, fuel, a battery |
| **Magnetic** | magnets attract or repel | two fridge magnets held apart |
| **Electrostatic** | charges attract or repel | a charged balloon |
| **Nuclear** | of the particles in the nucleus | uranium fuel |

## FOUR WAYS TO TRANSFER ENERGY

| TRANSFERRED… | WHAT HAPPENS | EXAMPLE |
|---|---|---|
| **mechanically** | a force moves something | pushing a trolley; a sound wave |
| **electrically** | a current flows | a battery lighting a lamp |
| **by heating** | a hotter object warms a colder one | a hob heating a pan |
| **by radiation** | light or infrared carries it | the Sun warming your face |

! **Stores hold energy. Transfers move it.** Light, sound and electricity are **not** stores: they are ways energy travels from one store to another.

## SYSTEMS

A **system** is the object, or group of objects, that we choose to study. In a **closed system**, no energy gets in or out, so the **total** energy stays the same. The energy only moves between stores.

## DESCRIBE ANY CHANGE IN THREE STEPS

1. Which store **empties**?
2. Which store **fills**?
3. **How** is the energy transferred: mechanically, electrically, by heating or by radiation?

**EXAMPLE: A CAR BRAKING**

The kinetic store of the car empties. The thermal store of the brakes fills. Friction is a force, so the energy is transferred mechanically. **Write:** “Energy is transferred mechanically from the kinetic store of the car to the thermal store of the brakes.”

PRACTISE

## TRY IT

**1** Draw a line from each store to its example.

| | |
|---|---|
| Kinetic | a stretched bow |
| Gravitational potential | a hot bath |
| Elastic potential | a rolling ball |
| Chemical | a battery |
| Thermal | a book on a high shelf |

**2** Draw a ring around the correct answer in each box.

Light is **an energy store** | **a way energy is transferred**

A battery stores energy in its **chemical store** | **kinetic store** | **thermal store**

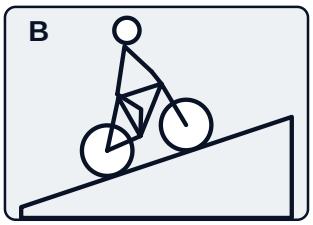
At the top of its flight, a ball has most energy in its

**kinetic store** | **gravitational potential store**

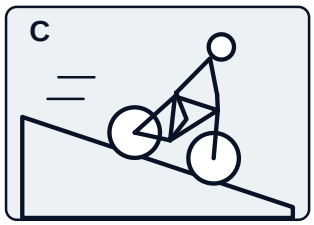
**3** The pictures tell a story. For each picture, write the store that **empties** and the store that **fills**. Choose from the words under each gap.



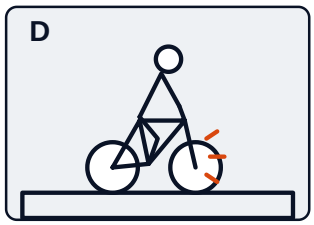

**Eating breakfast**




**Cycling up the hill**




**Freewheeling down**




**Braking at the bottom**

| PICTURE | STORE THAT EMPTIES | STORE THAT FILLS |
|---|---|---|
| **A Eating** | chemical (in the food) | chemical (in the body) |
| **B Cycling up** | **chemical / kinetic** | **gravitational potential / elastic potential** |
| **C Freewheeling down** | **gravitational potential / chemical** | **kinetic / nuclear** |
| **D Braking** | **kinetic / thermal** | **thermal / chemical** |

**4** SUPPORTED Complete each sentence. The choices are after each gap.

A ball is thrown upwards. Energy is transferred from the _____ **(kinetic / chemical)** store to the _____ **(gravitational potential / thermal)** store.

A moving car hits a wall. Energy is transferred from the _____ **(kinetic / elastic potential)** store of the car to the _____ **(thermal / chemical)** store of the car, the wall and the surroundings.

A kettle boils water. Energy is transferred _____ **(electrically / by radiation)** to the _____ **(thermal / kinetic)** store of the water.

**5** ON YOUR OWN A bungee jumper falls, and the cord stretches until the jumper stops at the bottom. Describe the energy transfers. Use the three steps.

**6** Explain what is wrong with what Sam says.

**Sam** When the car stopped, its energy was used up.

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** Name both stores and say which one empties and which one fills.

**7** A student throws a ball straight up. Complete the sentence. **[2 marks]**

As the ball rises, energy is transferred from the _____ **(kinetic / chemical / thermal)** store of the ball to its _____ **(elastic potential / gravitational potential / nuclear)** store.

**8** What is the unit of energy? Tick **one** box. **[1 mark]**

☐ joule ☐ newton ☐ watt ☐ kilogram

**9** Describe the energy transfer when an electric kettle heats water. **[2 marks]**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- Energy is measured in **joules (J)**. It is never made or destroyed.
- Stores: kinetic, gravitational potential, elastic potential, thermal, chemical, magnetic, electrostatic, nuclear.
- Transfers: mechanically, electrically, by heating, by radiation.
- In a closed system, the total energy stays the same.

### COVER THE PAGE AND ANSWER

**1** What is the unit of energy?

**2** Name four energy stores.

**3** Is light a store or a way of transferring energy?

**4** What is a closed system?

### I CAN…

- name the energy stores
- name the four ways energy is transferred
- explain what a system is
- describe energy transfers using “from … to …”

Lesson

# 02 THE KINETIC STORE

AQA Physics 4.1.1.2; Combined Science 6.1.1.2

**BIG QUESTION** **A car travels twice as fast. Does it have twice as much energy in its kinetic store, or more?**

**BY THE END YOU CAN**

- say what the kinetic store depends on
- calculate kinetic energy with $E_k = \frac{1}{2} m v^2$
- convert grams to kilograms
- explain why doubling the speed gives four times the energy

**kinetic store** **mass (kg)** **speed (m/s)** **squared**

**YOU WILL NEED**

Calculator. Practical: trolley, 1.00 m track, stopwatch, balance.



REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

Which store does a moving object have energy in?

**2**

A car brakes. Which store fills?

**3**

Light is _____ **(a store / a way energy is transferred).**



EXPLAIN

## ENERGY IN A MOVING OBJECT

Anything that moves has energy in its **kinetic store**. The more **mass** it has, and the faster it moves, the more energy it has.

$$E_k = \frac{1}{2} m v^2$$

**kinetic energy = 0.5 × mass × (speed)²**

$E_k$ in joules (J), $m$ in kilograms (kg), $v$ in metres per second (m/s)

Only the speed is squared. In this unit you use the equation to calculate the energy.

! **Square the speed first:** $v^2 = v \times v$. Then multiply by the mass and by 0.5.

**WORKED EXAMPLE**

**A 1200 kg car moves at 15 m/s. Calculate the energy in its kinetic store.**

| | |
|---|---|
| KNOW | m = 1200 kg, v = 15 m/s |
| EQUATION | $E_k = \frac{1}{2} m v^2$ |
| SUBSTITUTE | $E_k = 0.5 \times 1200 \times 15^2$ |
| ANSWER | $E_k = 135\,000$ J |

**YOUR TURN**

**A 60 kg runner moves at 5.0 m/s. Calculate the energy in their kinetic store.**

| | |
|---|---|
| KNOW | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | |

## MASS IN GRAMS? CONVERT FIRST

The mass must be in kilograms. Divide grams by 1000: $450\ \text{g} = 0.450\ \text{kg}$. A 450 g football kicked at 20 m/s has $E_k = 0.5 \times 0.450 \times 20^2 = 90$ J.

## DOUBLE THE SPEED, FOUR TIMES THE ENERGY

| SPEED OF A 1200 KG CAR | $v^2$ | $E_k$ |
|---|---|---|
| **15 m/s** | 225 | 135 000 J |
| **30 m/s (2 × faster)** | 900 | 540 000 J (4 × more) |
| **45 m/s (3 × faster)** | 2025 | 1 215 000 J (9 × more) |

Twice the speed gives $2^2 = 4$ times the energy. Three times the speed gives $3^2 = 9$ times. This is why a crash at high speed does so much more damage, and why the brakes must transfer so much more energy to stop the car. Doubling the **mass** only doubles the energy, because the mass is not squared.

PRACTISE

# TRY IT

**1** Draw a ring around the correct answer in each box.

Doubling the mass makes the kinetic energy

| | | |
|---|---|---|
| **stay the same** | **2 times bigger** | **4 times bigger** |

Doubling the speed makes the kinetic energy

| | | |
|---|---|---|
| **stay the same** | **2 times bigger** | **4 times bigger** |

**2** SUPPORTED A 2.0 kg ball rolls at 3.0 m/s. Calculate the energy in its kinetic store.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**3** SUPPORTED A 450 g football is kicked at 20 m/s. Calculate the energy in its kinetic store. *Convert the mass first!*

| | |
|---|---|
| CONVERT | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

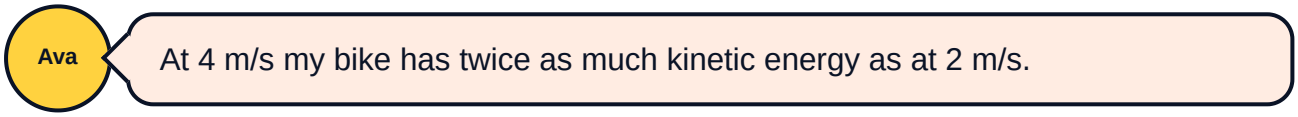
**4** Complete the table. Show your working underneath.

| Object | Mass | Speed | $E_k$ / J |
|---|---|---|---|
| **Lorry** | **30 000 kg** | **2.0 m/s** | |
| **Cyclist and bike** | **80 kg** | **6.0 m/s** | |
| **Tennis ball** | **58 g** | **50 m/s** | |
| **Bullet** | **10 g** | **400 m/s** | |

5 **ON YOUR OWN** A 0.16 kg cricket ball is bowled at 40 m/s. Calculate the energy in its kinetic store.

6 **CHALLENGE** A car speeds up from 10 m/s to 30 m/s. How many times bigger is the energy in its kinetic store? Explain without calculating the energy.

7 Explain what is wrong with what Ava says.



INVESTIGATE

# MEASURE THE ENERGY OF A MOVING TROLLEY

Work in a group of three: one pushes, one times, one records.

1. Measure the trolley's mass on the balance in grams. Convert it to kilograms.
2. Mark a 1.00 m section of the track with tape.
3. Give the trolley a gentle push so it rolls through the section. Time how long it takes to cross the 1.00 m.
4. Repeat twice more with the same push, and find the mean time.
5. Calculate the speed, then the kinetic energy.

**8** Record your results.

| Mass / g | Mass / kg | Time 1 / s | Time 2 / s | Time 3 / s | Mean time / s |
|---|---|---|---|---|---|
| | | | | | |

**9** Calculate the speed: $\text{speed} = \dfrac{\text{distance}}{\text{time}}$.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**10** Calculate the energy in the trolley's kinetic store.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**11** Which measurement has the biggest effect on your answer? **the mass** | **the time**

Explain why.

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** Write the equation, then substitute, then give the answer with its unit.

**12** A cyclist and bike have a total mass of 70 kg and move at 8.0 m/s. Calculate the energy in their kinetic store. **[2 marks]**

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**13** Car A and car B have the same mass. Car B moves twice as fast as car A. Complete the sentence. **[2 marks]**

Car B has _____ **(2 / 4 / 8)** times as much energy in its kinetic store as car A, because in the equation the speed is **halved** | **squared** | **doubled**.

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- $E_k = \frac{1}{2} m v^2$: mass in kg, speed in m/s, energy in J.
- Convert grams to kilograms: divide by 1000.
- Only the speed is squared: 2 × the speed gives 4 × the energy.
- 2 × the mass gives 2 × the energy.

### COVER THE PAGE AND ANSWER

**1** What two things does the kinetic store depend on?

**2** Write the equation for kinetic energy.

**3** Convert 250 g to kilograms.

**4** The speed triples. What happens to the kinetic energy?

### I CAN…

- calculate kinetic energy with $E_k = \frac{1}{2} m v^2$
- convert grams to kilograms
- explain why doubling the speed gives four times the energy

Lesson

# 03 THE GRAVITATIONAL POTENTIAL STORE

AQA Physics 4.1.1.2; Combined Science 6.1.1.2

**BIG QUESTION** **Carrying a bag up the stairs is hard work. Where does that energy go, and can you get it back?**

**BY THE END YOU CAN**

- explain what $g = 9.8$ N/kg means
- calculate energy with $E_p = m\,g\,h$
- use the height gained
- rearrange to find a height or a mass

**gravitational potential store**

**gravitational field strength** **newton (N)**

**height gained**

**ELECTRIC MOUNTAIN**

At Dinorwig in Wales, water is pumped up into a lake high on a mountain. When the country suddenly needs electricity, the water rushes down through turbines. The energy waits in the water's gravitational potential store until it is needed.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

In $E_k = \frac{1}{2}\,m\,v^2$, what does $v$ stand for? Give its unit.

**2**

Convert 600 g to kilograms.

**3**

A 2.0 kg ball moves at 4.0 m/s. Calculate its kinetic energy.

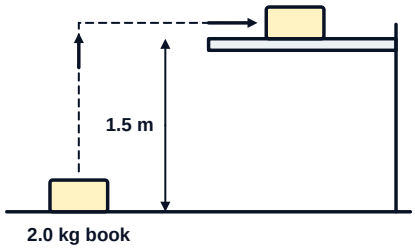
EXPLAIN

## ENERGY FROM HEIGHT

Lift an object up and you transfer energy to its **gravitational potential store**. Let it fall and the energy is transferred to its kinetic store. The heavier the object and the higher you lift it, the more energy it stores.

## WHAT $g$ MEANS

! On Earth, gravity pulls every kilogram down with a force of **9.8 newtons**. We write this as $g = 9.8$ N/kg (newtons per kilogram). It is called the **gravitational field strength**. On the Moon, $g = 1.6$ N/kg, so the same lift stores less energy there.

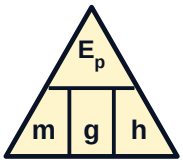



$$E_p = m\,g\,h$$

**gravitational potential energy = mass × gravitational field strength × height**

$E_p$ in J, $m$ in kg, $g$ in N/kg, $h$ in m

Rearranged: $h = \dfrac{E_p}{m\,g}$ and $m = \dfrac{E_p}{g\,h}$




### WORKED EXAMPLE

**A 2.0 kg book is lifted onto a shelf 1.5 m high. Calculate the energy transferred to its gravitational potential store.**

| | |
|---|---|
| KNOW | m = 2.0 kg, g = 9.8 N/kg, h = 1.5 m |
| EQUATION | $E_p = m\,g\,h$ |
| SUBSTITUTE | $E_p = 2.0 \times 9.8 \times 1.5$ |
| ANSWER | $E_p = 29.4$ J |

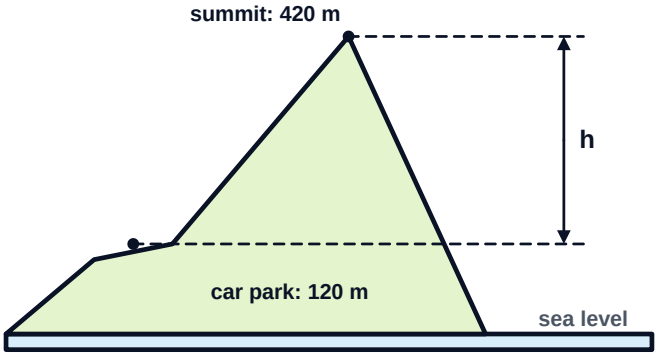
### YOUR TURN

**A 5.0 kg bag is lifted 2.0 m. Calculate the energy transferred to its gravitational potential store.**

| | |
|---|---|
| KNOW | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | |

## USE THE HEIGHT GAINED

Only the **change** in height matters. A hiker climbs from a car park 120 m above sea level to a summit 420 m above sea level. The height gained is $h = 420 - 120 = 300$ m. A 55 kg hiker gains $E_p = 55 \times 9.8 \times 300 = 161\,700$ J.



## FINDING THE HEIGHT OR THE MASS

**WORKED EXAMPLE**

**A 60 kg climber gains 17 640 J in their gravitational potential store. How high did they climb?**

| | |
|---|---|
| KNOW | $E_p = 17\,640$ J, m = 60 kg, g = 9.8 N/kg |
| REARRANGE | $h = \frac{E_p}{m\,g}$ |
| SUBSTITUTE | $h = \frac{17\,640}{60 \times 9.8}$ |
| ANSWER | **h = 30 m** |

**YOUR TURN**

**A crate is lifted 2.0 m and gains 392 J. Calculate its mass.**

KNOW

REARRANGE

SUBSTITUTE

ANSWER

**Get the marks:** Work out the bottom of the fraction first ($60 \times 9.8 = 588$), then divide. Or put brackets round it on your calculator.

PRACTISE

## TRY IT

**1** Draw a ring around the correct answer in each box.

On the Moon, lifting a box 1 m stores

| | | |
|---|---|---|
| **more energy than on Earth** | **less energy than on Earth** | **the same energy as on Earth** |

Lifting a box twice as high makes the energy stored

| | | |
|---|---|---|
| **half as big** | **twice as big** | **four times as big** |

**2** SUPPORTED A 0.50 kg ball is lifted 3.0 m. Calculate the energy transferred to its gravitational potential store.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**3** **SUPPORTED** A 600 g book is lifted 1.2 m onto a shelf. Calculate the energy transferred. *Convert the mass first!*

| | |
|---|---|
| CONVERT | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**4** Complete the table. Use $g = 9.8$ N/kg. Show your working underneath.

| Object | Mass / kg | Height gained / m | $E_p$ / J |
|---|---|---|---|
| **Child at the top of a slide** | **30** | **2.5** | |
| **Rock on a cliff** | **4.0** | | **196** |
| **Crate on a lorry** | | **3.0** | **882** |
| **Lift with passengers** | **800** | **12** | |

**5** **CHALLENGE** A 50 kg student gains 4.9 kJ in their gravitational potential store climbing stairs. How high did they climb? *Convert kJ to J first.*

INVESTIGATE

## ACTIVITY: LIFT IT

Work in pairs. Lift gently and keep the floor clear.

Choose three objects. Measure the mass of each and the height of the shelf or bench you lift it onto. Calculate the energy transferred to each gravitational potential store.

**6** Record and calculate.

| Object | Mass / kg | Height / m | $E_p$ / J |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

**7** How high would you need to lift your lightest object so that it stores the same energy as your heaviest object?

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** $g$ is always given in the question. AQA uses 9.8 N/kg.

**8** A crane lifts a 250 kg load through a height of 12 m. Calculate the energy transferred to the load's gravitational potential store. Gravitational field strength = 9.8 N/kg **[2 marks]**

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**9** A 0.40 kg ball gains 7.84 J in its gravitational potential store when it is thrown upwards. Calculate how high it rises. Gravitational field strength = 9.8 N/kg **[3 marks]**

| | |
|---|---|
| EQUATION | |
| REARRANGE | |
| SUBSTITUTE | |
| ANSWER | unit: |

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- $g = 9.8$ N/kg: gravity pulls each kilogram down with 9.8 N.
- $E_p = m\,g\,h$, with the height gained in metres.
- Rearranged: $h = \dfrac{E_p}{m\,g}$ and $m = \dfrac{E_p}{g\,h}$.
- Twice the height or twice the mass gives twice the energy.

### COVER THE PAGE AND ANSWER

**1** What does $g = 9.8$ N/kg mean?

**2** Write the equation for gravitational potential energy.

**3** A 5.0 kg bag is lifted 2.0 m. How much energy does it gain?

**4** Rearrange $E_p = m\,g\,h$ to find $h$.

### I CAN…

- explain what $g = 9.8$ N/kg means
- calculate $E_p = m\,g\,h$ using the height gained
- rearrange $E_p = m\,g\,h$ to find a height or a mass

Lesson

# 04 STRETCHING SPRINGS

AQA Physics 4.5.3; Combined Science 6.5.3

**BIG QUESTION** **A trampoline spring stretches a lot when you land on it. A car's suspension spring hardly moves when you sit in the car. What makes one spring stiffer than another?**

### BY THE END YOU CAN

- explain why two forces are needed to stretch a spring
- tell elastic and inelastic deformation apart
- use $F = ke$ to find a force, a spring constant or an extension
- plot two graphs of the same results and find the spring constant

**extension** **spring constant (N/m)** **elastic deformation** **inelastic deformation** **limit of proportionality**

### YOU WILL NEED

Clamp stand, spring, mass hanger, 100 g masses, metre rule, pointer, safety goggles.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

In $E_p = mgh$, what does $g$ stand for? Give its value on Earth.

**2**

A 2.0 kg box is lifted 3.0 m. Calculate the energy transferred to its gravitational potential store.

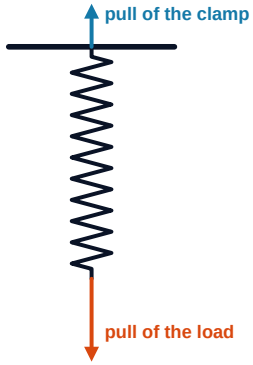
**3**

Convert 12 cm to metres.

EXPLAIN

# STRETCH, SQUASH, BEND

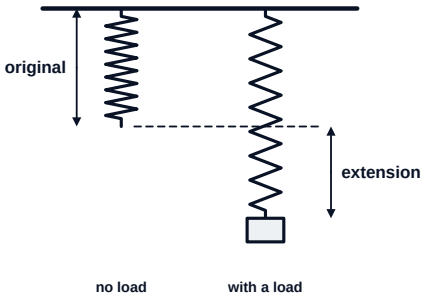
To stretch a spring and keep it still, you need **two forces**, one pulling each end. Here the clamp pulls the top up, and the load pulls the bottom down. With only one force, the spring would just move.




**Elastic deformation:** the spring goes back to its original length when the force is removed.
**Inelastic deformation:** the spring stays stretched.

## EXTENSION

> ! **extension = stretched length − original length**
> Always record the **extension**, not the length.



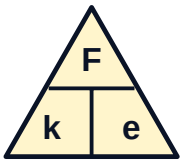

If a spring is 5.0 cm long with no load and 9.0 cm long with a load, its extension is 9.0 − 5.0 = 4.0 cm.

$$F = k\,e$$

**force = spring constant × extension**

$F$ in newtons (N), $k$ in newtons per metre (N/m), $e$ in metres (m)

Rearranged: $k = \frac{F}{e}$ and $e = \frac{F}{k}$




The **spring constant** $k$ tells you how stiff a spring is: the force needed for each metre of extension. A stiffer spring has a bigger $k$. In Lesson 5 you will use $k$ to work out how much energy a stretched spring stores.

> ! **Hooke's law:** the extension of a spring is directly proportional to the force on it, as long as the **limit of proportionality** is not passed. Double the force, double the extension.

**WORKED EXAMPLE**

**A spring stretches by 4.0 cm when a force of 2.0 N acts on it. Calculate its spring constant.**

| | |
|---|---|
| KNOW | F = 2.0 N, e = 4.0 cm = 0.040 m |
| REARRANGE | $k = \frac{F}{e}$ |
| SUBSTITUTE | $k = \frac{2.0}{0.040}$ |
| ANSWER | **k = 50 N/m** |

**YOUR TURN**

**A spring with k = 40 N/m is stretched by 0.15 m. Calculate the force.**

KNOW

EQUATION

SUBSTITUTE

ANSWER

## MASSES INTO FORCES AT THE BENCH

! Gravity pulls each kilogram down with 9.8 N, so each 100 g pulls down with 0.98 N. At the bench we round this: **each 100 g pulls down with about 1 N**. A mass is not a force: always record the force in newtons.

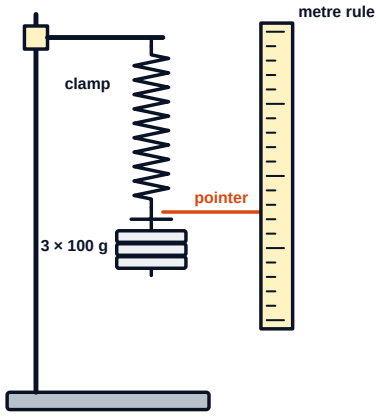
INVESTIGATE

## PRACTICAL: STRETCHING A SPRING

Work in pairs. Stand up while you work.

**STAY SAFE**

- Wear safety goggles: a spring that slips off can whip.
- Keep your feet out from under the masses.
- Add masses gently. Do not let them drop.



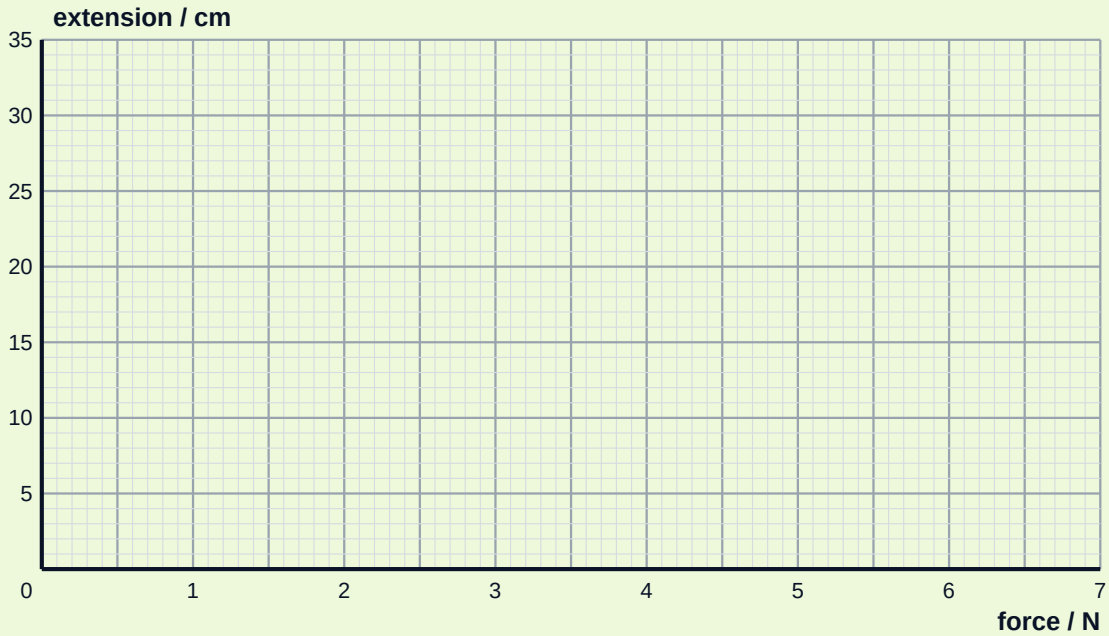

1. Line the pointer up with the metre rule. Record the original length.
2. Add a 100 g mass. Record the new length.
3. Repeat in 100 g steps up to 700 g.
4. Work out the force and the extension for each load.
5. Take the masses off one at a time. Does the spring go back to its original length?

6 Record your results. The force is about 1 N for each 100 g.

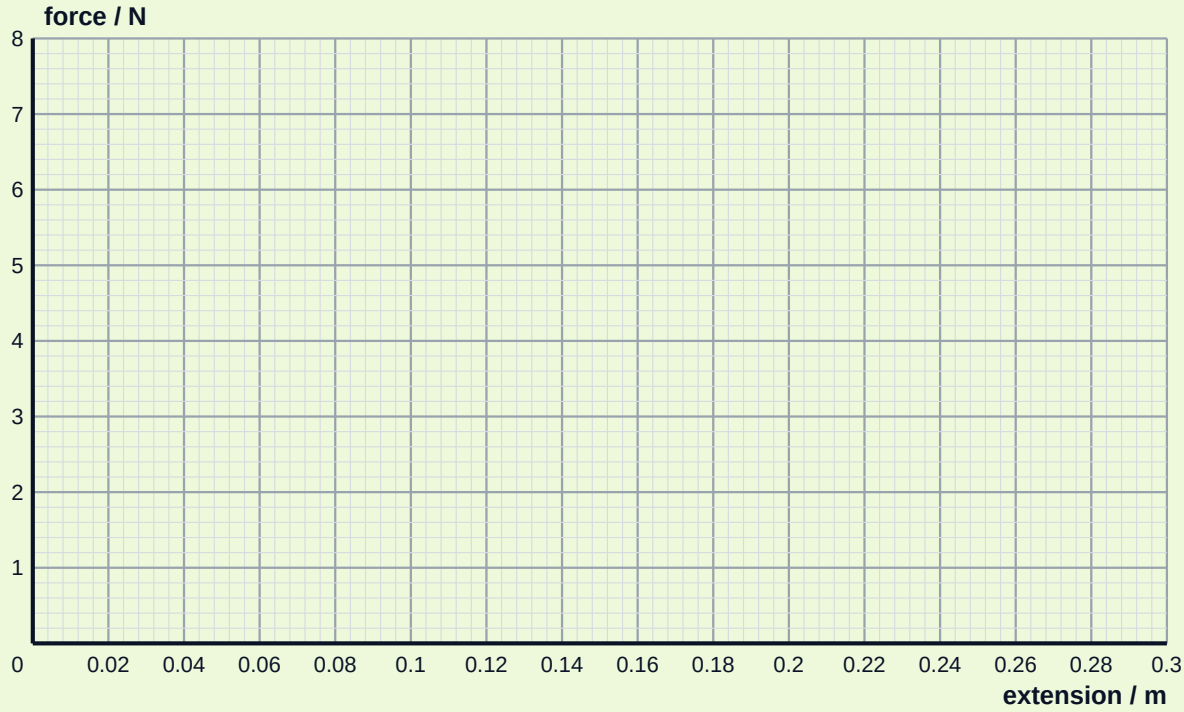
| Mass added / g | Force / N | Length / cm | Extension / cm |
|---|---|---|---|
| 0 | | | |
| 100 | | | |
| 200 | | | |
| 300 | | | |
| 400 | | | |
| 500 | | | |
| 600 | | | |
| 700 | | | |

## TWO GRAPHS OF THE SAME RESULTS

7 **Graph 1: extension against force.** Force is the independent variable (you chose it), so it goes on the x-axis. Plot the points and draw a best-fit line.



**8** **Graph 2: force against extension.** Swap the axes. Convert the extensions to metres first. Plot the points and draw a best-fit line.




**9** Use graph 2 to find the spring constant. On graph 2, $k$ is the gradient:

$$k = \frac{\text{change in force}}{\text{change in extension}}.$$

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**10** Both graphs show the same results. Complete the table to explain the difference.

| | Y-AXIS | X-AXIS | THE GRADIENT TELLS YOU |
|---|---|---|---|
| **Graph 1** | extension | force | how stretchy the spring is / how stiff the spring is |
| **Graph 2** | force | extension | how stretchy the spring is / how stiff the spring is |

**11** A stiffer spring is tested. Draw a ring around the correct answer in each box.

On graph 1, its line would be **steeper** | **less steep**

On graph 2, its line would be **steeper** | **less steep**

12 Which reading is beyond the limit of proportionality? How can you tell from the graphs?

PRACTISE

## TRY IT

1 Draw a ring around the correct answer in each box.

Elastic deformation means the object | **goes back to its original shape** | **stays stretched** |

A spring with a bigger spring constant is | **stiffer** | **easier to stretch** |

2 SUPPORTED A spring is 6.0 cm long. With a load on it, it is 9.5 cm long. Calculate the extension.

| | | |
|---|---|---|
| EQUATION | | |
| SUBSTITUTE | | |
| ANSWER | | unit: |

3 SUPPORTED Calculate the force needed to stretch a spring with k = 200 N/m by 0.050 m.

| | | |
|---|---|---|
| EQUATION | | |
| SUBSTITUTE | | |
| ANSWER | | unit: |

4 ON YOUR OWN A force of 3.0 N stretches a spring by 10 cm. Calculate its spring constant.

**5** **ON YOUR OWN** A spring with k = 40 N/m has a force of 2.0 N on it. Calculate the extension. Give your answer in cm.

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** Convert the extension to metres before you divide.

**13** A force of 5.0 N stretches a spring by 25 cm. Calculate the spring constant of the spring. **[3 marks]**

| | |
|---|---|
| CONVERT | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**14** Describe what happens to the extension of a spring when the force on it is doubled. Assume the limit of proportionality is not passed. **[2 marks]**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- Two forces are needed to stretch, squash or bend an object.
- Extension = stretched length − original length.
- $F = ke$: $k$ in N/m, $e$ in metres.
- On a force–extension graph (force on the y-axis), the gradient is $k$.
- Hooke's law holds up to the limit of proportionality.

### COVER THE PAGE AND ANSWER

**1** What is the extension of a spring?

**2** Write the equation linking force, spring constant and extension.

**3** What is the unit of spring constant?

**4** What does inelastic deformation mean?

### I CAN…

- explain why two forces are needed to stretch a spring
- calculate an extension from two lengths
- use $F = ke$ and its rearrangements
- plot both graphs and find k from the gradient

Lesson

# 05 THE ELASTIC POTENTIAL STORE

AQA Physics 4.1.1.2, 4.5.3; Combined Science 6.1.1.2, 6.5.3

**BIG QUESTION** **An archer pulls a bowstring back twice as far. Does the arrow get twice as much energy, or more?**

**BY THE END YOU CAN**

- calculate elastic potential energy with $E_e = \frac{1}{2}\,k\,e^2$
- find $k$ first when you are given a force
- explain why doubling the extension gives four times the energy
- say when the equation can be used

**elastic potential store** **work done** **compression**

**BUNGEE**

A bungee cord is a giant elastic band. As the jumper falls, the cord stretches and fills its elastic potential store. That slows the jumper down safely, then springs them back up.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

A spring has k = 50 N/m. Calculate the force that stretches it by 0.10 m.

**2**

What is the unit of spring constant?

**3**

Doubling the speed makes the kinetic energy _____ **(2 / 4 / 8)** times bigger.

EXPLAIN

## ENERGY IN A STRETCHED SPRING

When you stretch or squash a spring, you **do work** on it: your force transfers energy to it. The energy is stored in the spring's **elastic potential store**. Let go, and the spring gives the energy back.

$$E_e = \frac{1}{2}\,k\,e^2$$

**elastic potential energy = 0.5 × spring constant × (extension)²**

$E_e$ in J, $k$ in N/m, $e$ in metres (m)

Only the extension is squared. For a squashed spring, $e$ is how much shorter it is. In this unit you use the equation to calculate the energy.

! The equation works only up to the **limit of proportionality**. Beyond it, the spring may be permanently stretched and will not give all the energy back.

**WORKED EXAMPLE**

**A spring with k = 50 N/m is stretched by 0.040 m. Calculate the energy in its elastic potential store.**

| | |
|---|---|
| KNOW | k = 50 N/m, e = 0.040 m |
| EQUATION | $E_e = \frac{1}{2}\,k\,e^2$ |
| SUBSTITUTE | $E_e = 0.5 \times 50 \times 0.040^2$ |
| ANSWER | $E_e = 0.04$ J |

**YOUR TURN**

**A spring with k = 200 N/m is squashed by 0.050 m. Calculate the energy stored.**

KNOW

EQUATION

SUBSTITUTE

ANSWER

## GIVEN A FORCE? FIND $k$ FIRST

**WORKED EXAMPLE**

**A force of 3.0 N stretches a spring by 10 cm. Calculate the energy stored.**

| | |
|---|---|
| KNOW | F = 3.0 N, e = 10 cm = 0.10 m |
| STEP 1 | $k = \frac{F}{e} = \frac{3.0}{0.10} = 30$ N/m |
| STEP 2 | $E_e = \frac{1}{2}\,k\,e^2 = 0.5 \times 30 \times 0.10^2$ |
| ANSWER | $E_e = 0.15$ J |

**YOUR TURN**

**A force of 4.0 N stretches a spring by 0.20 m. Calculate the energy stored.**

KNOW

STEP 1

STEP 2

ANSWER

## DOUBLE THE STRETCH, FOUR TIMES THE ENERGY

| EXTENSION OF A 50 N/M SPRING | $e^2$ | $E_e$ |
|---|---|---|
| **0.040 m** | 0.0016 | 0.04 J |
| **0.080 m (2 × further)** | 0.0064 | 0.16 J (4 × more) |
| **0.120 m (3 × further)** | 0.0144 | 0.36 J (9 × more) |

This is the same pattern as kinetic energy, because the extension is squared. **Why?** The further you stretch a spring, the harder you have to pull. So each extra centimetre takes more work than the one before.

PRACTISE

## TRY IT

**1** Draw a ring around the correct answer in each box.

$E_e = \frac{1}{2} k e^2$ can only be used up to the

| | | |
|---|---|---|
| **limit of proportionality** | **top of the ruler** | **point where the spring snaps** |

Tripling the extension makes the energy stored

| | | |
|---|---|---|
| **3 times bigger** | **6 times bigger** | **9 times bigger** |

**2** SUPPORTED A spring with k = 80 N/m is stretched by 0.10 m. Calculate the energy stored.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**3** **SUPPORTED** A car's suspension spring has k = 400 N/m and is squashed by 20 mm. Calculate the energy stored. *Convert mm to m: divide by 1000.*

| | |
|---|---|
| CONVERT | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**4** **ON YOUR OWN** A force of 6.0 N stretches a spring by 0.15 m. Calculate the energy stored.

**5** Raj's spring has k = 30 N/m. What did he do wrong? Calculate the correct answer.

> **Raj:** e = 10 cm, so the energy stored is $0.5 \times 30 \times 10^2 = 1500$ J.

**6** **CHALLENGE** Spring A has k = 100 N/m and is stretched by 0.10 m. Spring B has k = 50 N/m and is stretched by 0.20 m. Which spring stores more energy? Show your working.

INVESTIGATE

## YOUR OWN SPRING

Use the spring constant you found in Lesson 4.

**7** My spring constant from Lesson 4: $k$ = _____ N/m. Calculate the energy stored at each extension.

| Extension / m | $e^2$ / m² | $E_e$ / J |
|---|---|---|
| **0.05** | | |
| **0.10** | | |
| **0.20** | | |

**8** Check: did doubling the extension from 0.10 m to 0.20 m give four times the energy?

**yes** | **no** Show how you know.

**9** Could you use $E_e = \frac{1}{2} k e^2$ for the 700 g reading in Lesson 4? **yes** | **no**

Explain.

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** Square only the extension. Give the unit.

**10** A catapult has a spring constant of 500 N/m. It is stretched by 0.12 m. Calculate the energy in its elastic potential store. **[2 marks]**

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**11** Explain why $E_e = \frac{1}{2}\,k\,e^2$ cannot be used once the spring has passed its limit of proportionality. **[2 marks]**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- $E_e = \frac{1}{2}\,k\,e^2$: $k$ in N/m, $e$ in metres, $E_e$ in J.
- The work done stretching a spring is stored in its elastic potential store.
- Given a force and an extension? Find $k = \dfrac{F}{e}$ first.
- 2 × the extension gives 4 × the energy.
- Only up to the limit of proportionality.

### COVER THE PAGE AND ANSWER

**1** Write the equation for elastic potential energy.

**2** What unit must the extension be in?

**3** The extension doubles. What happens to the energy stored?

**4** When can the equation not be used?

### I CAN…

- calculate $E_e = \frac{1}{2}\,k\,e^2$ with the extension in metres
- find $k$ from a force first, then calculate $E_e$
- explain why doubling the extension gives four times the energy

Lesson

# 06 CONSERVATION OF ENERGY

AQA Physics 4.1.2.1, 4.1.1.1; Combined Science 6.1.2.1, 6.1.1.1

**BIG QUESTION** **A heavy pendulum is pulled back to touch your nose and let go. When it swings back, will it hit you?**

**BY THE END YOU CAN**

- state the principle of conservation of energy
- follow energy from one store to another in a closed system
- use conservation of energy in calculations
- draw energy bar charts
- explain why real systems lose some useful energy

**conservation of energy** **closed system** **dissipated** **energy bar chart**

**YOU WILL NEED**

Practical: clamp stand, string, pendulum bob, metre rule, a phone to film the swing.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

A 1.0 kg ball is lifted 2.0 m. Calculate the energy transferred to its gravitational potential store.

**2**

A 2.0 kg ball moves at 3.0 m/s. Calculate the energy in its kinetic store.

**3**

A stretched spring stores energy in its _____ **(kinetic / elastic potential)** store.

EXPLAIN

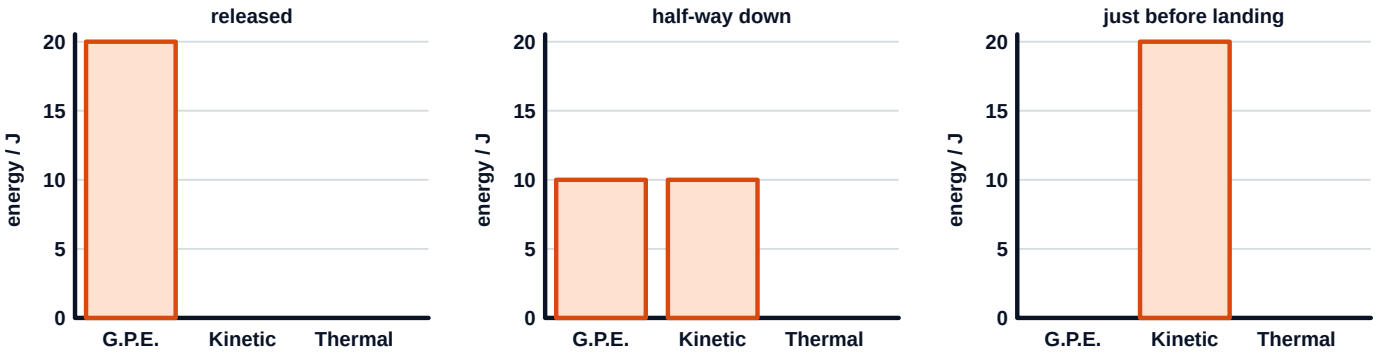
## ENERGY IS NEVER LOST

! **Conservation of energy:** energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.

In a **closed system**, the total energy never changes. Energy only moves from one store to another.

## A FALLING BALL

As a ball falls, energy is transferred from its gravitational potential (G.P.E.) store to its kinetic store. If we ignore air resistance, all the energy that leaves the G.P.E. store arrives in the kinetic store. An **energy bar chart** shows this: the bars change, but the total stays the same.




**WORKED EXAMPLE**

**A 0.50 kg ball is dropped from 3.0 m. Calculate the energy in its kinetic store just before it lands. Ignore air resistance.**

**IDEA**

energy gained by the kinetic store = energy lost from the G.P.E. store

**EQUATION**

$$E_p = m\,g\,h$$

**SUBSTITUTE**

$$E_p = 0.50 \times 9.8 \times 3.0 = 14.7 \text{ J}$$

**ANSWER**

$$E_k = 14.7 \text{ J}$$

**YOUR TURN**

**A 2.0 kg rock falls 5.0 m. Calculate the energy in its kinetic store just before it lands. Ignore air resistance.**

**IDEA**

**EQUATION**

**SUBSTITUTE**

**ANSWER**

## FINDING A HEIGHT

**WORKED EXAMPLE**

**A 0.25 kg ball is thrown straight up with 20 J in its kinetic store. Calculate the greatest height it reaches. Ignore air resistance.**

**IDEA** at the top, all 20 J is in the G.P.E. store

**REARRANGE** $h = \frac{E_p}{m\,g}$

**SUBSTITUTE** $h = \frac{20}{0.25 \times 9.8}$

**ANSWER** **h = 8.2 m**

**YOUR TURN**

**A 0.10 kg ball is thrown straight up with 4.9 J in its kinetic store. How high does it go? Ignore air resistance.**

**IDEA**

**REARRANGE**

**SUBSTITUTE**

**ANSWER**

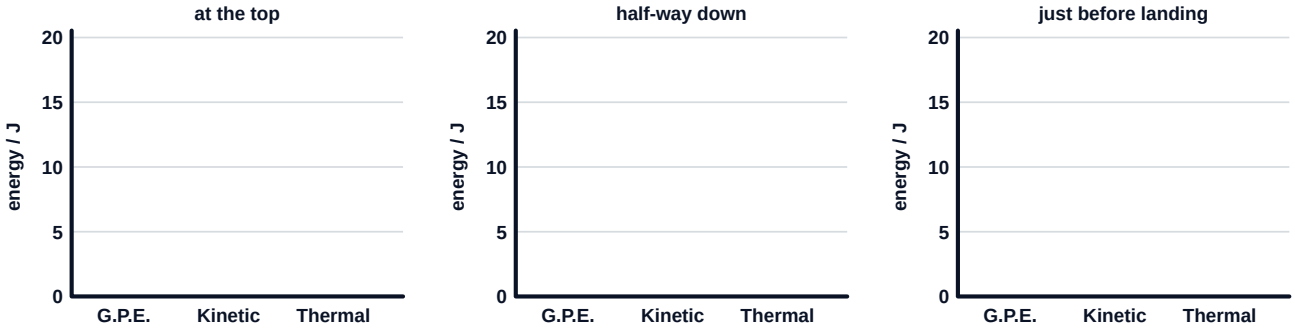
## REAL SYSTEMS: SOME ENERGY IS DISSIPATED

In real life, air resistance and friction transfer some energy to the **thermal store** of the surroundings. This energy is not destroyed, but it is spread out and we cannot use it again. We say it is **dissipated**. That is why a pendulum swings a little lower each time, and why it will not hit your nose.

PRACTISE

### TRY IT

1 A ball has 20 J in its G.P.E. store at the top. Ignore air resistance. Draw the bars at each moment.



**2** Complete the sentences. The choices are after each gap.

At the top of its swing, a pendulum has most energy in its _____ **(kinetic / gravitational potential)** store.

At the bottom of its swing, it has most energy in its _____ **(kinetic / gravitational potential)** store.

The total energy _____ **(goes up / goes down / stays the same)** if we ignore air resistance.

**3** **ON YOUR OWN** A 0.20 kg apple falls 2.5 m from a tree. Calculate the energy in its kinetic store just before it lands. Ignore air resistance.

**4** **CHALLENGE** A toy launcher has a spring with k = 200 N/m, squashed by 0.050 m. It fires a 10 g ball straight up. Calculate the greatest height the ball could reach.

**5** Explain why Mo is wrong.

**Mo:** The pendulum swings lower each time, so energy is being destroyed.

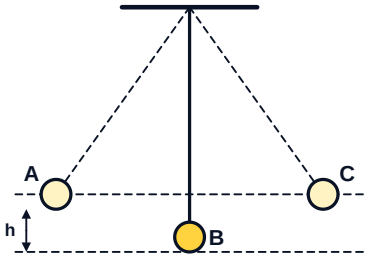
INVESTIGATE

# PRACTICAL: DOES THE PENDULUM KEEP ITS ENERGY?

Every reading is a height, so you do not need fast timing.

**STAY SAFE**

- Clamp the stand firmly so it cannot tip.
- Let the bob go from rest; never push it.
- Keep clear of the swing.




1. Hang the bob. Note the height of its lowest point on the metre rule.
2. Pull the bob to the side until it is 0.20 m higher than its lowest point (A). Let go.
3. Read the height it reaches on the other side (C). Filming helps.
4. Let it keep swinging. Read the height it reaches after 10 swings.

6. Mass of the bob = _____ kg. Record the heights and calculate the energy in the G.P.E. store each time.

| Moment | Height above lowest point / m | $E_p$ / J |
|---|---|---|
| **Released** | **0.20** | |
| **After 1 swing** | | |
| **After 10 swings** | | |

7. On the first swing, how much energy is in the bob's kinetic store at the lowest point B? Explain.

8 After 10 swings, the energy has been **created** | **destroyed** | **dissipated** . Where has it gone?

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** In part (b), say why the two energies are equal.

9 A diver of mass 60 kg stands on a board 10 m above the water. Gravitational field strength = 9.8 N/kg. **[2 marks]**

(a) Calculate the energy in the diver's gravitational potential store.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

10 (b) The diver falls into the water. Ignoring air resistance, how much energy is in the diver's kinetic store just before they reach the water? Give a reason. **[2 marks]**

11 (c) In real life, the energy in the kinetic store is a little less. Explain why. **[2 marks]**

REVIEW LATER

# REVISE IT

Use this box after the lesson and before tests.

## SUMMARY

- Energy cannot be created or destroyed, only transferred, stored or dissipated.
- In a closed system the total energy stays the same.
- Falling: G.P.E. store → kinetic store. Rising: kinetic store → G.P.E. store.
- Energy bar charts: the bars change, the total stays the same.
- Real systems dissipate some energy to the thermal store of the surroundings.

## COVER THE PAGE AND ANSWER

**1** State the principle of conservation of energy.

**2** A ball falls and loses 12 J of G.P.E. Ignoring air resistance, how much does its kinetic store gain?

**3** Why does a pendulum swing lower each time?

**4** What does an energy bar chart show?

## I CAN…

- state the principle of conservation of energy
- use energy lost = energy gained in calculations
- draw energy bar charts
- explain why real systems dissipate energy

Lesson

# 07 POWER

AQA Physics 4.1.1.4; Combined Science 6.1.1.4

**BIG QUESTION** **Two students carry the same bag up the same stairs. One runs and one walks. Who transferred more energy? Who was more powerful?**

**BY THE END YOU CAN**

- say what power means
- state that 1 watt is 1 joule per second
- calculate power with $P = \frac{E}{t}$, and find $E$ or $t$
- convert kW to W, kJ to J and minutes to seconds

**power** **watt (W)** **kilowatt (kW)** **rate**

**YOU WILL NEED**

Practical: stopwatch, tape measure or metre rule, a flight of stairs.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

A 50 kg student climbs 3.0 m of stairs. Calculate the energy transferred to their G.P.E. store.

**2**

Convert 2.0 minutes to seconds.

**3**

State the principle of conservation of energy.

EXPLAIN

## HOW FAST IS ENERGY TRANSFERRED?

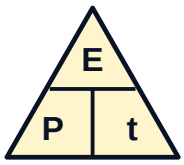
**Power** is how fast energy is transferred: the energy transferred **each second**. Two motors can lift the same load through the same height, so they transfer the same energy. The motor that does it faster is more powerful.

$$P = \frac{E}{t}$$

**power = energy transferred ÷ time**

$P$ in watts (W), $E$ in joules (J), $t$ in seconds (s)

Rearranged: $E = Pt$ and $t = \frac{E}{P}$




**!** **1 watt (W) = 1 joule per second.** 1 kilowatt (kW) = 1000 W. A 60 W lamp transfers 60 J every second.

| | ENERGY TRANSFERRED | TIME | POWER |
|---|---|---|---|
| **Motor A** | 29.4 J | 2.0 s | 29.4 ÷ 2.0 = 14.7 W |
| **Motor B** | 29.4 J | 6.0 s | 29.4 ÷ 6.0 = 4.9 W |

**WORKED EXAMPLE**

**A motor transfers 600 J in 4.0 s. Calculate its power.**

KNOW E = 600 J, t = 4.0 s

EQUATION $P = \frac{E}{t}$

SUBSTITUTE $P = \frac{600}{4.0}$

ANSWER **P = 150 W**

**YOUR TURN**

**A lamp transfers 900 J in 15 s. Calculate its power.**

KNOW

EQUATION

SUBSTITUTE

ANSWER

**WORKED EXAMPLE**

**A 2.0 kW kettle is switched on for 2.0 minutes. Calculate the energy transferred.**

CONVERT P = 2000 W, t = 120 s

REARRANGE $E = Pt$

SUBSTITUTE $E = 2000 \times 120$

ANSWER **E = 240 000 J**

**YOUR TURN**

**A 1500 W motor must transfer 45 000 J. How long does it take?**

KNOW

REARRANGE

SUBSTITUTE

ANSWER

PRACTISE

# TRY IT

**1** Draw a ring around the correct answer in each box.

Power is measured in | **joules** | **watts** | **newtons** |

A more powerful motor transfers energy | **faster** | **slower** |

1 W is the same as | **1 J per second** | **1 J per minute** | **1 N per metre** |

**2** SUPPORTED A 40 W fan runs for 5.0 minutes. Calculate the energy transferred. *Convert the time first!*

| | |
|---|---|
| CONVERT | |
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**3** Complete the table. Show your working underneath.

| Appliance | Energy / J | Time / s | Power / W |
|---|---|---|---|
| **Fan** | **1200** | **30** | |
| **Kettle** | | **60** | **2000** |
| **Lamp** | **5000** | | **100** |

**4** CHALLENGE A 3.0 kW heater transfers 540 kJ. How many minutes was it on for?

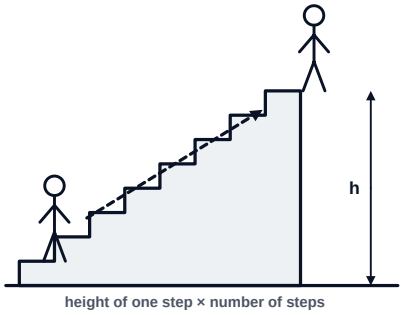
INVESTIGATE

# PRACTICAL: HOW POWERFUL ARE YOU?

Everyone uses a mass of 50 kg, so nobody is weighed.

**STAY SAFE**

- Walk briskly. Do not run. Hold the handrail.
- One person on the stairs at a time.
- Use dry, clear stairs.




1. Measure the height of one step. Count the steps. Height h = step height × number of steps.
2. Your partner times you walking briskly up the stairs.
3. Calculate the energy transferred: $E = m\,g\,h$, with $m = 50$ kg.
4. Calculate your power: $P = \frac{E}{t}$.

| | |
|---|---|
| **Height of one step / m** | |
| **Number of steps** | |
| **Height h / m** | |
| **Mass / kg** | 50 |
| **Time / s** | |

5. Calculate the energy transferred to your G.P.E. store.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

6. Calculate your power.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**7** Is your real power output **higher** | **lower** than your answer? Explain.

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** Two steps: find the energy first, then the power.

**8** A lift motor raises a 600 kg lift through 15 m in 20 s. Gravitational field strength = 9.8 N/kg. **[2 marks]**

(a) Calculate the energy transferred to the lift's gravitational potential store.

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**9** (b) Calculate the useful power output of the motor. **[2 marks]**

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**10** Motor A lifts a load in 5 s. Motor B lifts the same load through the same height in 10 s. Complete the sentence. **[2 marks]**

Motor A transfers _____ **(more / less / the same)** energy as motor B, and has _____ **(a greater / a smaller / the same)** power.

REVIEW LATER

# REVISE IT

Use this box after the lesson and before tests.

## SUMMARY

- Power is the rate of energy transfer: the energy transferred each second.
- $P = \frac{E}{t}$; $E = Pt$; $t = \frac{E}{P}$.
- 1 W = 1 J/s; 1 kW = 1000 W.
- Time in seconds: minutes × 60.

## COVER THE PAGE AND ANSWER

**1** What is power?

**2** What is 1 watt?

**3** A lamp transfers 600 J in 10 s. What is its power?

**4** Write the equation to find energy from power and time.

## I CAN…

- define power as the rate of energy transfer
- calculate $P = \frac{E}{t}$ and find $E$ or $t$
- convert kW, kJ and minutes
- measure my own power

Lesson

# 08 WASTED ENERGY

AQA Physics 4.1.2.1; Combined Science 6.1.2.1

**BIG QUESTION** **A hot chocolate always goes cold. Where does the energy go, and how can you slow it down?**

**BY THE END YOU CAN**

- explain what “dissipated” means and where the energy ends up
- explain how lubrication reduces unwanted energy transfers
- explain how thermal conductivity and wall thickness affect how fast a house cools
- suggest and explain ways to insulate a home

**dissipated** **friction** **lubrication** **conduction** **thermal conductivity** **thermal insulator**

**SEEING HEAT**

A thermal camera shows warm things as bright colours. Point one at a house in winter and the windows, the roof and the gaps round the door glow: that is where energy is escaping into the cold air outside.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

A fan transfers 600 J in 20 s. Calculate its power.

**2**

Dissipated energy usually ends up in the _____ **(thermal / chemical / nuclear)** store of the surroundings.

**3**

A pendulum swings lower each time. Why?

EXPLAIN

## REDUCING UNWANTED ENERGY TRANSFERS

In every real energy transfer, some energy goes to stores where it is not useful, usually the **thermal store of the surroundings**. This energy is **dissipated** (spread out). People often call it “wasted” energy. It is not destroyed, but we cannot use it again.

| UNWANTED TRANSFER | HOW TO REDUCE IT | WHY IT WORKS |
|---|---|---|
| **Friction between moving parts** | Lubricate with oil or grease | The surfaces slide past each other more easily, so less energy goes to the thermal store |
| **Air resistance on a vehicle** | A streamlined shape | The air pushes back less, so less energy goes to the thermal store of the air |
| **Conduction through walls and roofs** | Thermal insulation | Materials with a low thermal conductivity slow the transfer |

## CONDUCTION AND THERMAL CONDUCTIVITY

Energy is transferred through a solid by **conduction**, from the hotter side to the colder side. A material with a **higher thermal conductivity** transfers energy **faster**. Metals have a high thermal conductivity. Air, wool and plastic foam have a low thermal conductivity, so they are good **thermal insulators**. Many insulators work because they trap air.

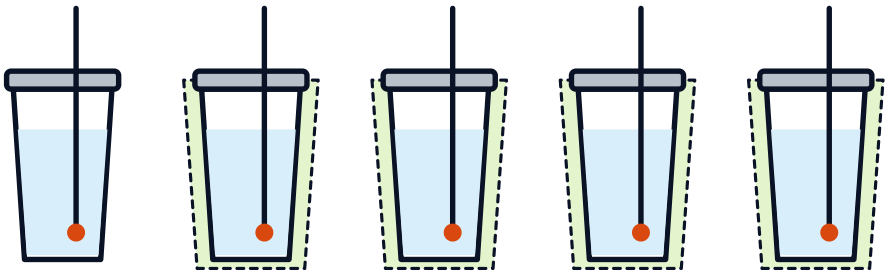
> ! A house cools **more slowly** if its walls are **thicker** and have a **lower thermal conductivity**.

INVESTIGATE

### PRACTICAL: COOLING CUPS

TRIPLE ONLY

Your teacher pours the hot water.




**Question:** which material is the best thermal insulator?

| | |
|---|---|
| **Independent variable** | |
| **Dependent variable** | |
| **Control variables (two)** | |

**STAY SAFE**

- The water is about 60 °C: hot enough to scald. Do not move the cups.
- Stand the cups in a tray. Report any spill straight away.

**5** Record your results.

| Material | Start / °C | After 10 min / °C | Drop / °C |
|---|---|---|---|
| **none** | | | |
| **newspaper** | | | |
| **bubble wrap** | | | |
| **cotton wool** | | | |
| **felt** | | | |

**6** The material is a categoric variable, so draw a **line graph** | **bar chart** of the temperature drop. Draw it here.

**7** Complete the conclusion.

The best insulator is the material whose cup had the _____ **(biggest / smallest)** temperature drop.

In my results, the best insulator was _____ **(none / newspaper / bubble wrap / cotton wool / felt)**.

Good insulators trap air, which has a _____ **(high / low)** thermal conductivity.

PRACTISE

## TRY IT

**1** Draw a line from each unwanted energy transfer to the best way to reduce it.

| | |
|---|---|
| Friction in a bike chain | a draught excluder |
| Air resistance on a lorry | oil on the chain |
| Conduction through a loft | a streamlined shape |
| Draughts under a door | thick loft insulation |

**2** Draw a ring around the correct answer in each box.

Copper has a high thermal conductivity, so copper is a good

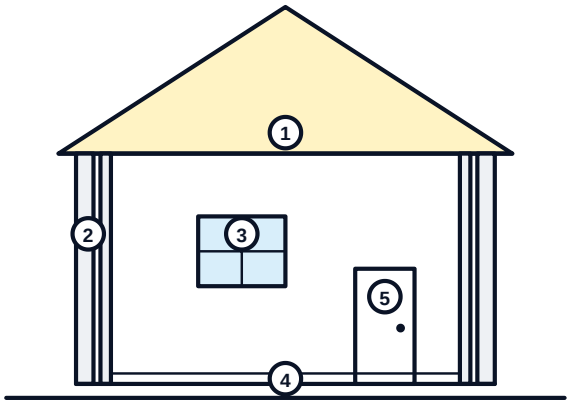
**thermal insulator** | **thermal conductor**

Making a wall twice as thick makes the house cool **faster** | **more slowly**

**3** Explain what insulation really does.

**Lily**: Insulation keeps the cold out.

**4** DESIGN **Design an Eco House.** For each numbered place on the house, name a feature that reduces energy transfer out of the house, and explain why it works. Use the words *thermal conductivity* or *thickness*.




| PLACE | FEATURE | WHY IT WORKS |
|---|---|---|
| **1 Roof** | | |
| **2 Walls** | | |
| **3 Window** | | |
| **4 Floor** | | |
| **5 Door** | | |

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** Say what the lubricant does to friction, and then what that does to the energy.

**8** Explain how oiling a bicycle chain reduces the energy wasted when you cycle. **[2 marks]**

9 Two houses have walls of the same thickness. The bricks of house A have a higher thermal conductivity than the bricks of house B. Which house cools faster? Explain. **[2 marks]**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- Dissipated (“wasted”) energy usually goes to the thermal store of the surroundings.
- Lubrication reduces friction, so less energy is dissipated.
- Higher thermal conductivity = faster energy transfer by conduction.
- Thicker walls with lower thermal conductivity make a house cool more slowly.
- Many insulators trap air.

### COVER THE PAGE AND ANSWER

**1** Where does dissipated energy usually end up?

**2** How does lubrication reduce wasted energy?

**3** What does a high thermal conductivity mean?

**4** Give two ways to make a house cool more slowly.

### I CAN…

- explain what dissipated means
- explain how lubrication reduces wasted energy
- explain the effect of thermal conductivity and wall thickness
- design and justify ways to insulate a home

Lesson

# 09 EFFICIENCY

AQA Physics 4.1.2.2; Combined Science 6.1.2.2

**BIG QUESTION** **An old filament lamp and an LED give out the same light. Why does the LED cost much less to run?**

**BY THE END YOU CAN**

- calculate efficiency as a decimal and as a percentage
- explain why no device is 100% efficient
- find the useful output or the total input from the efficiency
- draw and read Sankey diagrams
- describe ways to increase efficiency

**efficiency** **useful output** **wasted output** **total input** **Sankey diagram**

**LAMPS**

A filament lamp transfers only about 1 joule in every 10 by light. The rest heats the room. An LED transfers several times more of its energy by light, so it needs much less energy for the same brightness.

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**

A 2000 W heater is on for 30 s. Calculate the energy transferred.

**2**

Where does most wasted energy end up?

**3**

Give one way to reduce friction in a machine.

EXPLAIN

## HOW MUCH IS USEFUL?

**Efficiency** tells you what fraction of the energy put into a device is transferred usefully. The rest is wasted, usually dissipated to the thermal store of the surroundings.

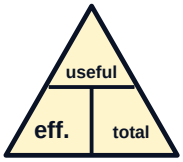
$$\text{efficiency} = \frac{\text{useful output energy transfer}}{\text{total input energy transfer}}$$

**Energy in joules (J). Efficiency has no unit.**

Answer as a decimal (for example 0.80), or multiply by 100 for a percentage (80%).

Rearranged: useful output = efficiency × total input, and

$$\text{total input} = \frac{\text{useful output}}{\text{efficiency}}$$




$$\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}}$$

**The same idea, using power in watts (W).**

Use whichever pair the question gives you: energies or powers.

! Efficiency is **never more than 1** (100%). If your answer is bigger than 1, you have divided the wrong way round. To use a percentage in a calculation, change it to a decimal first: 25% = 0.25.

### WORKED EXAMPLE

**A motor is supplied with 500 J and transfers 400 J to the kinetic store of its load. Calculate its efficiency.**

| | |
|---|---|
| KNOW | useful = 400 J, total = 500 J |
| EQUATION | $\text{efficiency} = \frac{\text{useful}}{\text{total}}$ |
| SUBSTITUTE | $\text{efficiency} = \frac{400}{500}$ |
| ANSWER | **0.80, or 80%** |

### YOUR TURN

**A lamp is supplied with 60 J and transfers 12 J by light. Calculate its efficiency.**

KNOW

EQUATION

SUBSTITUTE

ANSWER

**WORKED EXAMPLE**

**A 60 W lamp has an efficiency of 25%. Calculate its useful power output.**

| | |
|---|---|
| CONVERT | efficiency = 25% = 0.25 |
| REARRANGE | useful = efficiency × total |
| SUBSTITUTE | useful = 0.25 × 60 |
| ANSWER | **15 W** |

**YOUR TURN**

**A motor is 0.80 efficient. It transfers 400 J usefully. Calculate the total energy supplied.**

KNOW

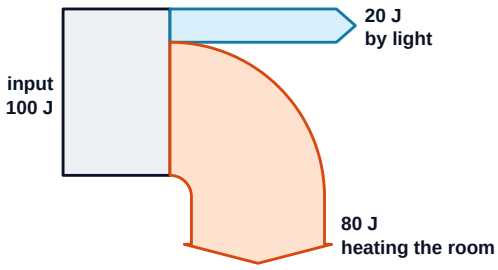
REARRANGE

SUBSTITUTE

ANSWER

## SANKEY DIAGRAMS

1. The **width** of each arrow shows the amount of energy.
2. Useful energy goes **straight on**. Wasted energy **bends away**, usually downwards.
3. Input = useful + wasted. Here: 100 J = 20 J + 80 J.
4. Efficiency = 20 ÷ 100 = 0.20 (20%).




A lamp: 100 J in, 20 J by light.

## INCREASING EFFICIENCY

Reduce the wasted energy: **lubricate** moving parts (less friction), **streamline** vehicles (less air resistance), **insulate** hot things (less energy transferred by heating), and choose devices that waste less, such as LEDs instead of filament lamps.

PRACTISE

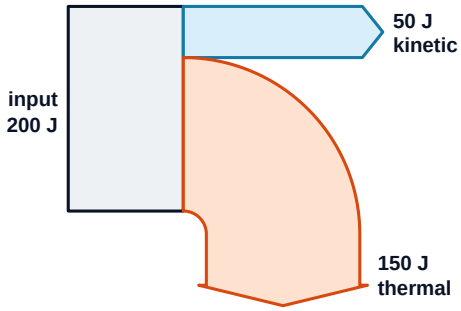
## TRY IT

1. Draw a ring around the correct answer in each box.

An efficiency of 1.4 is **possible** | **impossible**

Efficiency has the unit **joules** | **watts** | **no unit**

**2** This is the Sankey diagram for an electric motor. Calculate its efficiency.




| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**3** A kettle is supplied with 2000 J. It transfers 1800 J to the thermal store of the water, and 200 J heats the kettle and the room. Draw its Sankey diagram. Let 1 square = 100 J.

**4** Complete the table. Show your working underneath.

| Device | Total input | Useful output | Efficiency |
|---|---|---|---|
| TV | 200 J | 60 J | |
| Electric motor | 1000 J | | 0.75 |
| Solar panel | | 30 W | 0.15 |

**5** Complete the sentences. The choices are after each gap.

Oiling the gears of a machine _____ **(increases / decreases)** its efficiency, because less energy is transferred by _____ **(friction / light)**.

An LED is _____ **(more / less)** efficient than a filament lamp, because _____ **(more / less)** energy is wasted heating the room.

INVESTIGATE

## PRACTICAL: HOW EFFICIENT IS A BOUNCE?

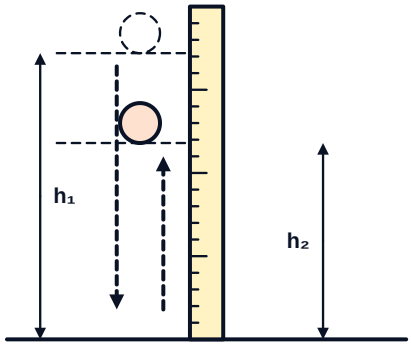
Filming the bounce against the ruler makes it much easier to read.

**1** Drop a ball from 1.00 m, measured from the bottom of the ball.

**2** Measure how high it bounces back (the bottom of the ball again).

**3** Do this three times and find the mean bounce height.

**4** Repeat with a different ball.




At the top of each bounce, the useful energy is all in the G.P.E. store. The mass and $g$ are the same both times, so:

$$\text{efficiency} = \frac{m\,g\,h_2}{m\,g\,h_1} = \frac{h_2}{h_1}$$

**6** Record your results.

| Ball | Drop height $h_1$ / m | Bounce 1 / m | Bounce 2 / m | Bounce 3 / m | Mean $h_2$ / m | Efficiency |
|---|---|---|---|---|---|---|
| | **1.00** | | | | | |
| | **1.00** | | | | | |

**7** Where did the rest of the energy go?

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** Check: is your efficiency less than 1 (100%)?

**8** A 1200 W hair dryer has a useful power output of 1080 W. Calculate its efficiency as a percentage. **[3 marks]**

| | |
|---|---|
| EQUATION | |
| SUBSTITUTE | |
| ANSWER | unit: |

**9** Explain why an LED is more efficient than a filament lamp. **[2 marks]**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- $\text{efficiency} = \dfrac{\text{useful output}}{\text{total input}}$, using energy or power.
- As a decimal (at most 1) or a percentage (at most 100%). No unit.
- useful = efficiency × total; total = useful ÷ efficiency.
- Sankey diagrams: arrow width shows energy; wasted energy bends away.
- Higher: lubricate, streamline, insulate to increase efficiency.

### COVER THE PAGE AND ANSWER

**1** Write the equation for efficiency using energy.

**2** A device transfers 30 J usefully from 120 J. What is its efficiency?

**3** Why can efficiency never be more than 100%?

**4** On a Sankey diagram, what does the width of an arrow show?

### I CAN…

- calculate efficiency as a decimal and a percentage
- find the useful output or the total input
- draw and read a Sankey diagram
- describe ways to increase efficiency

REFERENCE

# KEY WORDS

Learn the meaning and the spelling. In exams, precise words earn marks: say “transferred to the thermal store of the surroundings”, never “lost”.

**Energy store** L1: Where energy is held: kinetic, gravitational potential, elastic potential, thermal, chemical, magnetic, electrostatic or nuclear

**Joule (J)** L1: The unit of energy

**System** L1: The object or group of objects that we choose to study

**Closed system** L1: A system that no energy enters or leaves; its total energy stays the same

**Kinetic store** L2: Energy an object has because it is moving

**Gravitational potential store** L3: Energy an object has because it is raised in a gravitational field

**Gravitational field strength (g)** L3: The force of gravity on each kilogram: 9.8 N/kg on Earth

**Extension** L4: Stretched length minus original length

**Spring constant (k)** L4: The force needed for each metre of extension, in N/m; a stiffer spring has a bigger k

**Elastic deformation** L4: The object goes back to its original shape when the force is removed

**Inelastic deformation** L4: The object stays stretched or bent when the force is removed

**Limit of proportionality** L4: The point beyond which extension is no longer directly proportional to force

**Elastic potential store** L5: Energy an object has because it is stretched or squashed

**Conservation of energy** L6: Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed

**Dissipated** L6: Spread out to the surroundings, usually to the thermal store, where it is no longer useful

**Power** L7: The rate of energy transfer: energy transferred per second

**Watt (W)** L7: The unit of power: 1 W = 1 J/s

**Lubrication** L8: Oil or grease between moving parts, to reduce friction

**Thermal conductivity** L8: How fast a material transfers energy by conduction; higher means faster

**Thermal insulator** L8: A material with a low thermal conductivity

**Efficiency** L9: Useful output ÷ total input; a decimal up to 1 or a percentage up to 100%

**Sankey diagram** L9: A diagram in which the width of each arrow shows the amount of energy

FINISH

# PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

| I CAN… | R / A / G |
|---|---|
| **01: Energy stores and systems** | |
| name the energy stores | |
| name the four ways energy is transferred | |
| explain what a system is | |
| describe energy transfers using “from … to …” | |
| **02: The kinetic store** | |
| calculate kinetic energy with $E_k = \frac{1}{2} m v^2$ | |
| convert grams to kilograms | |
| explain why doubling the speed gives four times the energy | |
| **03: The gravitational potential store** | |
| explain what $g = 9.8$ N/kg means | |
| calculate $E_p = m g h$ using the height gained | |
| rearrange $E_p = m g h$ to find a height or a mass | |
| **04: Stretching springs** | |
| explain why two forces are needed to stretch a spring | |
| calculate an extension from two lengths | |
| use $F = k e$ and its rearrangements | |
| plot both graphs and find k from the gradient | |
| **05: The elastic potential store** | |
| calculate $E_e = \frac{1}{2} k e^2$ with the extension in metres | |
| find $k$ from a force first, then calculate $E_e$ | |
| explain why doubling the extension gives four times the energy | |
| **06: Conservation of energy** | |
| state the principle of conservation of energy | |
| use energy lost = energy gained in calculations | |

| I CAN… | R / A / G |
|---|---|
| draw energy bar charts | |
| explain why real systems dissipate energy | |
| **07: Power** | |
| define power as the rate of energy transfer | |
| calculate $P = \frac{E}{t}$ and find $E$ or $t$ | |
| convert kW, kJ and minutes | |
| measure my own power | |
| **08: Wasted energy** | |
| explain what dissipated means | |
| explain how lubrication reduces wasted energy | |
| explain the effect of thermal conductivity and wall thickness | |
| design and justify ways to insulate a home | |
| **09: Efficiency** | |
| calculate efficiency as a decimal and a percentage | |
| find the useful output or the total input | |
| draw and read a Sankey diagram | |
| describe ways to increase efficiency | |

CHECK YOURSELF

# QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

## LESSON 1: ENERGY STORES AND SYSTEMS

### DO NOW

1. For example, a moving car (it is moving) or a hot drink (it is hot). Any sensible answer.
2. joules (J)
3. Warmer: rubbing (friction) transfers energy to the thermal store of your hands.

### COVER THE PAGE

1. Joule (J).
2. Any four of: kinetic, gravitational potential, elastic potential, thermal, chemical, magnetic, electrostatic, nuclear.
3. A way of transferring energy (by radiation).
4. A system that no energy enters or leaves, so its total energy stays the same.

## LESSON 2: THE KINETIC STORE

### DO NOW

1. The kinetic store.
2. The thermal store (of the brakes and the surroundings).
3. A way energy is transferred (by radiation).

### COVER THE PAGE

1. Mass and speed.
2. $E_k = \frac{1}{2} m v^2$
3. 0.250 kg
4. It becomes 9 times bigger.

## LESSON 3: THE GRAVITATIONAL POTENTIAL STORE

### DO NOW

1. Speed, in metres per second (m/s).
2. 0.600 kg
3. $E_k = 0.5 \times 2.0 \times 4.0^2$
   $E_k = 16$ J

### COVER THE PAGE

1. Gravity pulls each kilogram down with a force of 9.8 N.
2. $E_p = m g h$
3. 98 J
4. $h = \dfrac{E_p}{m g}$

## LESSON 4: STRETCHING SPRINGS

### DO NOW

1. Gravitational field strength: 9.8 N/kg.
2. $E_p = 2.0 \times 9.8 \times 3.0$
   $E_p = 58.8$ J
3. 0.12 m

### COVER THE PAGE

1. Its stretched length minus its original length.
2. $F = k e$
3. Newtons per metre (N/m).
4. The object does not go back to its original shape when the force is removed.

## LESSON 5: THE ELASTIC POTENTIAL STORE

### DO NOW

1. $F = k e = 50 \times 0.10 = 5.0$ N
2. N/m
3. 4

### COVER THE PAGE

1. $E_e = \frac{1}{2} k e^2$
2. Metres (m).
3. It becomes 4 times bigger.
4. Beyond the limit of proportionality.

## LESSON 6: CONSERVATION OF ENERGY

### DO NOW

1. $E_p = 1.0 \times 9.8 \times 2.0$
   $E_p = 19.6$ J
2. $E_k = 0.5 \times 2.0 \times 3.0^2$
   $E_k = 9.0$ J
3. elastic potential

### COVER THE PAGE

1. Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed.
2. 12 J
3. Some energy is dissipated to the thermal store of the surroundings by air resistance and friction.
4. How much energy is in each store at one moment.

## LESSON 7: POWER

### DO NOW

1. $E_p = 50 \times 9.8 \times 3.0$
   $E_p = 1470$ J
2. 120 s
3. Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed.

### COVER THE PAGE

1. The rate of energy transfer (energy transferred per second).
2. 1 joule per second.
3. 60 W
4. $E = Pt$

## LESSON 8: WASTED ENERGY

### DO NOW

1. P = 600 ÷ 20 = 30 W
2. thermal
3. Some energy is dissipated by air resistance and friction.

### COVER THE PAGE

1. In the thermal store of the surroundings.
2. It reduces friction, so less energy is transferred to the thermal store.
3. The material transfers energy by conduction quickly.
4. Thicker walls; materials with a lower thermal conductivity (insulation).

## LESSON 9: EFFICIENCY

### DO NOW

1. E = P t = 2000 × 30 = 60 000 J
2. The thermal store of the surroundings.
3. Lubricate it with oil or grease.

### COVER THE PAGE

1. efficiency = useful output energy transfer ÷ total input energy transfer
2. 0.25 (25%)
3. Energy cannot be created: the useful output can never be more than the input.
4. The amount of energy.

REFERENCE

# EQUATION TOOLKIT

Every equation in the unit. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it. Use $g = 9.8\ \text{N/kg}$.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find $h$ from $E_p = m\,g\,h$, cover $h$. $E_p$ is above $m$ and $g$, so $h = \dfrac{E_p}{m\,g}$. Kinetic and elastic potential energy have no triangle: in this unit you only use them to calculate the energy.

## $E_k = \frac{1}{2}\,m\,v^2$

Lesson 2

**kinetic energy = 0.5 × mass × (speed)²**

J, kg, m/s

Use it to calculate the energy. Only the speed is squared.

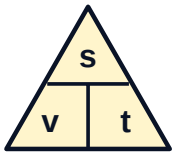
## $s = v\,t$

Lesson 2

**distance travelled = speed × time**

m, m/s, s

$v = \dfrac{s}{t}$ and $t = \dfrac{s}{v}$



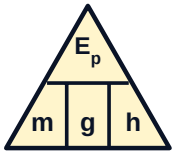

## $E_p = m\,g\,h$

Lesson 3

**gravitational potential energy = mass × gravitational field strength × height**

J, kg, N/kg, m

$h = \dfrac{E_p}{m\,g}$ and $m = \dfrac{E_p}{g\,h}$



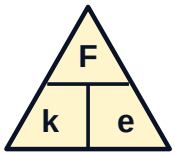

## $F = k\,e$

Lesson 4

**force = spring constant × extension**

N, N/m, m

$k = \dfrac{F}{e}$ and $e = \dfrac{F}{k}$




## $E_e = \frac{1}{2}\,k\,e^2$

Lesson 5

**elastic potential energy = 0.5 × spring constant × (extension)²**

J, N/m, m

Use it to calculate the energy. Only the extension is squared.

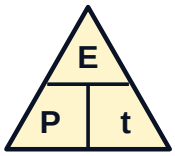
## $P = \dfrac{E}{t}$

Lesson 7

**power = energy transferred ÷ time**

W, J, s

$E = P\,t$ and $t = \dfrac{E}{P}$



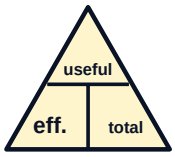

## $\text{eff.} = \dfrac{\text{useful output}}{\text{total input}}$

Lesson 9

**efficiency = useful output ÷ total input, using energy (J) or power (W)**

no unit: a decimal up to 1, or × 100 for a percentage

useful = eff. × total; $\text{total} = \dfrac{\text{useful}}{\text{eff.}}$




### 1

Write what you know. Convert first: g → kg, cm → m, kJ → J, min → s, % → decimal.

### 2

Write the equation in symbols.

### 3

Rearrange if you need to, then substitute the numbers.

### 4

Give the answer with a unit. Ask: is it sensible?